



A NEW ARC-LENGTH METHOD FOR HANDLING SHARP SNAP-BACKS

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Abstract—Various constraints for the arc-length method have been successfully employed to pass limit points. However, at very sharp snap-backs, particularly those occurring in fracture mechanics or damage mechanics both for failure initiation and crack propagation, standard arc-length methods can fail to converge and alternative solution procedures are required. A novel algorithm to choose the correct root for the cylindrical arc-length method is suggested here. It will be demonstrated that this new arc-length method is robust and can handle very sharp snap-backs. © 1998 Elsevier Science Ltd

INTRODUCTION

During the progressive failure analysis of a double cantilever beam (DCB) loaded under pure mode I, using interface elements employing a softening type damage model [1,2], it was found that the solutions often failed to converge beyond a critical displacement or load when the conventional cylindrical arc-length method [3,4] was used. The convergence problems frequently coincided with the crack initiation, or started in the initial stages of the crack propagation. Convergence problems of some sort persisted for nearly all simulations of DCB specimens. In some cases they disappeared for particular combinations of the finite element mesh and the energy release rate, but then reappeared for longer crack lengths. Characteristic of the convergence problems were oscillation cycles, where the iterative solutions circled infinitely between a number of trial solutions and the solution failed to converge. Problems with existing arc-length methods have been reported by a number of authors [5–7] and are often associated with snap-backs (which may be local). One successful solution [5] involves the use of the crack opening displacement to form the constraint equation. These procedures inevitably involve relative displacements between two (or more) nodes and consequently require pre-knowledge of the crack position and its local topology.

It might be possible to generalize such methods so as to limit this pre-knowledge, but the coupling between the resulting constraint relation and the element connectivity would still be inevitable. Particularly, once multiple cracks are considered, very considerable difficulties are anticipated. In

these circumstances, it might be better to revert to a method such as the cylindrical procedure [3], which automatically includes elements of constraint for all variables.

However, as previously discussed, in its present form the method is not very robust, particularly with sharp snap-backs. This paper will show that the main limitation is related to the choice of roots and hence it is possible to simply amend the method while retaining the main elements of the existing algorithms.

STANDARD ARC-LENGTH METHOD

To overcome limit points (Fig. 1), various constraint methods have been proposed (eg normal plane or updated normal plane method, the dominant displacement method, or the spherical or cylindrical arc length method [8–13]). In constraint methods, a load level parameter λ is introduced to adjust the load level. It is assumed that the problem involves the monotonic factoring of a fixed load vector $\mathbf{f}$ so that $\mathbf{f}^n = \lambda^n \mathbf{f}$. The constraint methods are effectively Newton Raphson procedures with an additional constraint to define the direction of the iterative path/surface. Many constraint equations have been proposed such as the spherical or the cylindrical arc length method [4], the normal plane or updated normal plane method [11] and the dominant displacement method [5,12].

In the standard method the iterative displacements which account for a flexible load level are given by:

$$\delta \mathbf{p}_i^n = \delta \bar{\mathbf{p}}_i^n + \delta \lambda_i^n \delta \hat{\mathbf{p}}_i^n \quad (1)$$

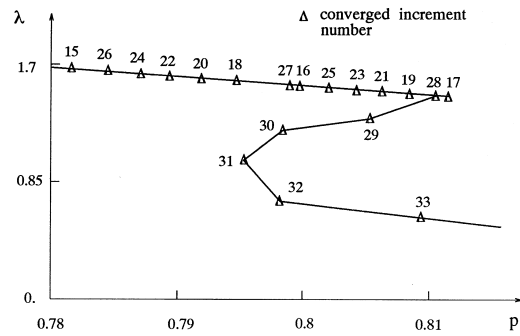


Fig. 4. Snap-back response during unstable crack growth.

length. A restart with a very small arc length in the proximity of a critical crack length showed an interesting phenomenon. The solution for a reversible material model doubled back on itself, returned, and then snapped back and continued to subsequently propagate. In Fig. 4 the converged increments for an extreme case of an unstable crack propagation are shown. It can be seen that before the solution followed the unstable path, it doubled back several times. With an elastic damage material model, we were not able to follow the unstable path, but were trapped in oscillation cycles. Figure 5 illustrates the iterative path for the reversible and elastic damage model at the critical increment (increment 18): whilst the reversible model doubles back on itself, the elastic damage model oscillates between the predictor and the first trial solution. Even though the reversible material model exhibited the remote possibility of reaching convergence, the standard arc-length method is clearly not satisfactory and alternative solution procedures are imperative.

Table 1. Residual at critical increment

Increment number	Root	Angle	Residual norm
18/1	a	30.58°	0.45273E + 04
18/1	b	118.63°	0.69244E - 03

NEW ARC-LENGTH METHOD

The convergence problems occurred when we selected the root corresponding to the smaller angle between the incremental displacements and the displacements from the predictor solution using Equation (7). In Fig. 6 the angles for both roots at the critical increment are plotted—the smaller angle α results in the doubling back, whereas with the bigger angle β we can immediately follow the unstable path, which snaps back sharply. Consequently, an alternative criterion for the root selection is required.

Clearly a better technique is required to choose the appropriate root in the presence of sharp snap-backs. The proposed method has its origin in Fig. 7, which indicates that the residual following the application of the “wrong solution” (although smallest angle), is likely to be much bigger than that associated with the “correct solution”. To apply this idea, it is necessary to try both alternatives, and update the displacements and compute the resulting residuals for both cases. The resulting overhead is not very large and is roughly the equivalent of the cost of one line-search. In relation to the previous example, Table 1 shows the data associated with the two separate roots at increment 18 (compare with Fig. 6). To save computer resources this more sophisticated procedure needs only to be switched on and activated when the user anticipates sharp snap-backs. In summary, the pro-

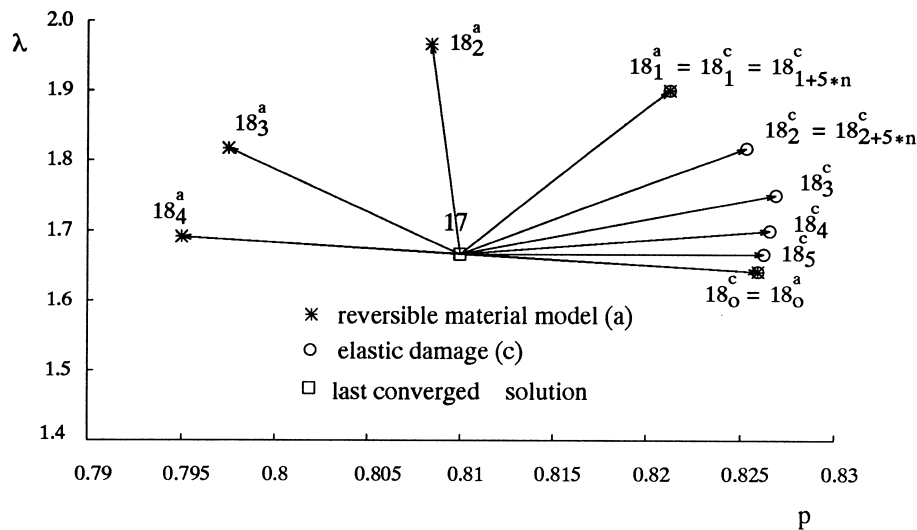


Fig. 5. Effect of the unloading mode on the iterative path during a snap-back.

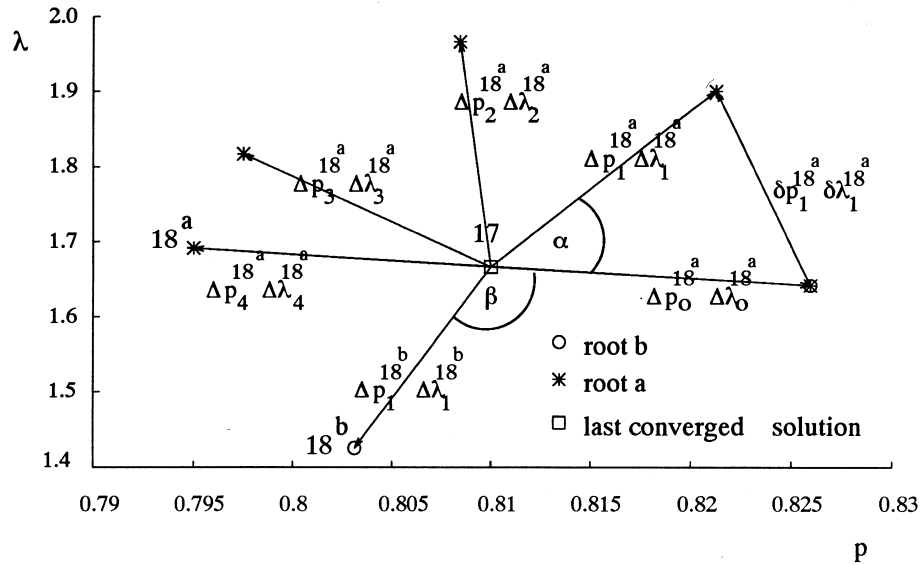


Fig. 6. Effect of the choice of roots on the iterative path (reversible un-/loading).

posed new root selection is based on trying both solutions and choosing the one with the minimum residual norm (either Euclidian or maximum). The steps are outlined in the following algorithm (here using the maximum norm).

1. standard arc length algorithm up to the computation of iterative vectors Equation (2 and 3)
2. solve quadratic equation for both roots $\delta\lambda_{i,j}^n$ Equation (4 and 6)

$$c_2 \delta\lambda_i^{n^2} + c_1 \delta\lambda_i^n + c_0 = 0$$

$$c_2 = \delta\hat{\mathbf{p}}_i^{n^t} \delta\hat{\mathbf{p}}_i^n$$

$$c_1 = 2\delta\hat{\mathbf{p}}_i^{n^t} (\Delta\hat{\mathbf{p}}_{i-1}^n + \delta\bar{\mathbf{p}}_i^n)$$

$$c_0 = \Delta\hat{\mathbf{p}}_{i-1}^{n^t} \Delta\hat{\mathbf{p}}_{i-1}^n + 2\Delta\hat{\mathbf{p}}_{i-1}^{n^t} \bar{\mathbf{p}}_i^n + \delta\bar{\mathbf{p}}_i^{n^t} \delta\bar{\mathbf{p}}_i^n - \Delta l^2$$

store $\delta\lambda_{i,j}^n$ $j = 1, 2$ set $j = 1$

3. update incremental load level
 $\Delta\lambda_{i,j}^n = \Delta\lambda_{i-1}^n + \delta\lambda_{i,j}^n$ and total load level
 $\lambda^n = \lambda^{n-1} + \Delta\lambda_{i,j}^n$
4. compute iterative displacements Equation (1)
5. update current total displacements, strains, stresses and internal forces
6. compute residual $\mathbf{g}_{i,j}^n = \lambda^n \mathbf{f} + \mathbf{q}_{i,j}^n$

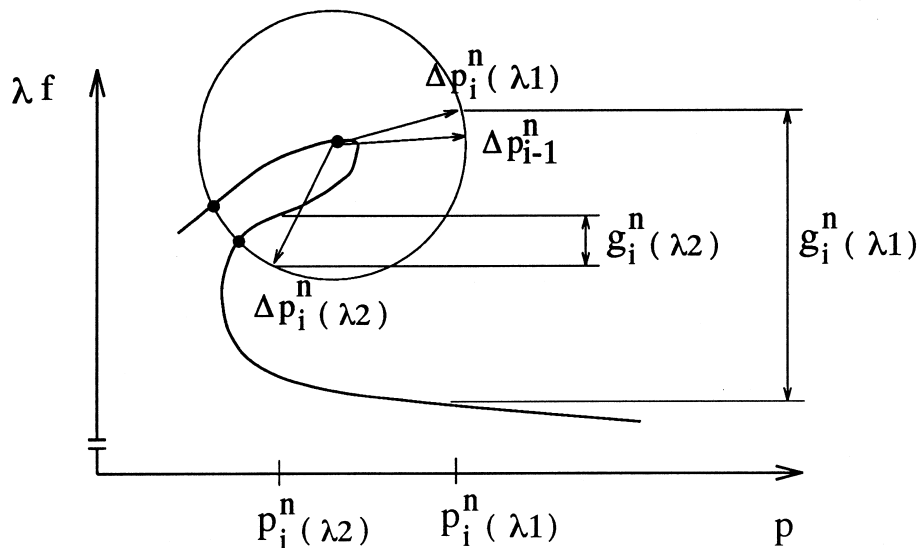


Fig. 7. Illustration of the minimum residual criterion.

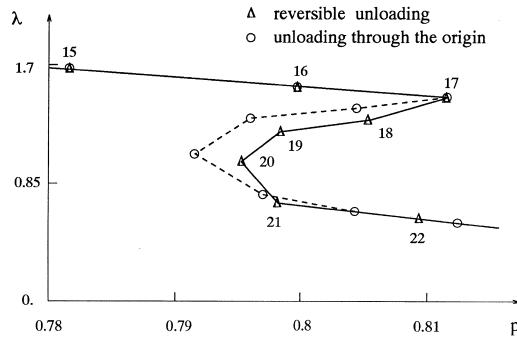


Fig. 8. Snap-back response revisited—modified choice of roots.

7. if the solution has converged, continue with next increment ($n + 1$)

8. if the solution has not converged

(a) if $j = 1$ set $j = 2$ and continue with 3.

(b) if $j = 2$ take $\delta\lambda_{i,j}^n$ and $\delta\mathbf{p}_{i,j}^n$ for $\min[\text{abs}(\mathbf{g}_{i,1}^n); \text{abs}(\mathbf{g}_{i,2}^n)]$, update the incremental load level and the incremental displacements and continue with next iteration ($i + 1$).

By selecting the root with the minimum residual, we can follow the load displacement curve continuously for both material unloading modes (Fig. 8).

SUMMARY AND CONCLUSIONS

The cylindrical and spherical arc-length methods suffer from convergence problems. This applies particularly to local instabilities in the form of snap-backs. The convergence problems are connected to an incorrect choice of roots for the quadratic constraint equation. In conventional methods, the root which gives the smallest angle between the previous and the current incremental displacements is selected. It has been shown in the paper that a more robust and efficient criterion is the minimum residual. This new criterion can easily be added to an existing arc-length method by including one additional loop over the number of roots. With only the additional overhead of approximately one line-search, paths with very sharp snap-backs can now be followed successfully.

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